THAI ROAD ACCIDENTS & SAFETY ANALYSIS REPORT – Group 5

DSA 1080A

Group Project By :

Austin Isinta - 677045

Imani Moronge - 677322

Mark Kimathi - 676294

Caleb Murai - 676502

Shallom Omufila - 676176

Valerie Achieng - 675945

Nicole Serem - 675791

Isaac Mwangi - 676296

Introduction

This report applies advanced data science techniques to analyze road accident data, uncovering patterns and predictive insights.

Road safety remains a global concern. This study leverages data analytics to evaluate accident patterns.

Methodology

Data cleaning, transformation, exploratory data analysis, and visualization techniques were applied using Python, Matplotlib, Seaborn and Folium

Results

Multiple trends were identified across time, location, and fatalities.

Discussion

Temporal and spatial patterns indicate systemic risk factors affecting road safety.

Insights

1. Speeding and driving under the influence of alcohol appear to be the most frequent causes of accidents. These human factors are the main causes of accidents in Thailand. (as from 2019-2022)
2. Many accidents happen on straight roads, despite wide curves and slopes.
3. A significant number of accidents happen during clear weather conditions.
4. Motorcycles are involved in a higher percentage of road accidents than private passenger cars.
5. With more than 81,000 documented incidents, the dataset highlights that road accidents are a widespread, systemic problem that calls for comprehensive, data-driven solutions rather than isolated, local interventions.

Recommendations

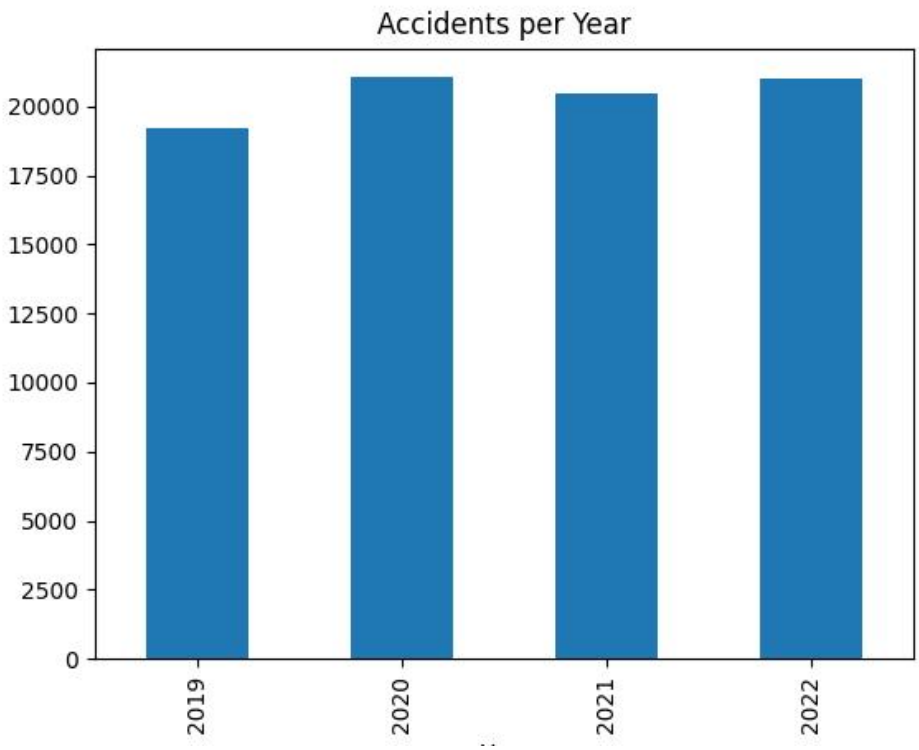
1. Focus sobriety checkpoints in high-accident provinces like Chiang Mai and Bangkok, particularly during late-night hours when alcohol-related accidents are most likely to occur.
2. Automated speed cameras and average speed zones should be installed and established on long straight road sections. This is due to the high percentage of road accidents on straight roads.
3. Motorcycle lanes could be established in urban provinces to ensure there is space for the vulnerable riders from the larger passenger vehicles.

Conclusion

Data-driven insights can guide policy and interventions.

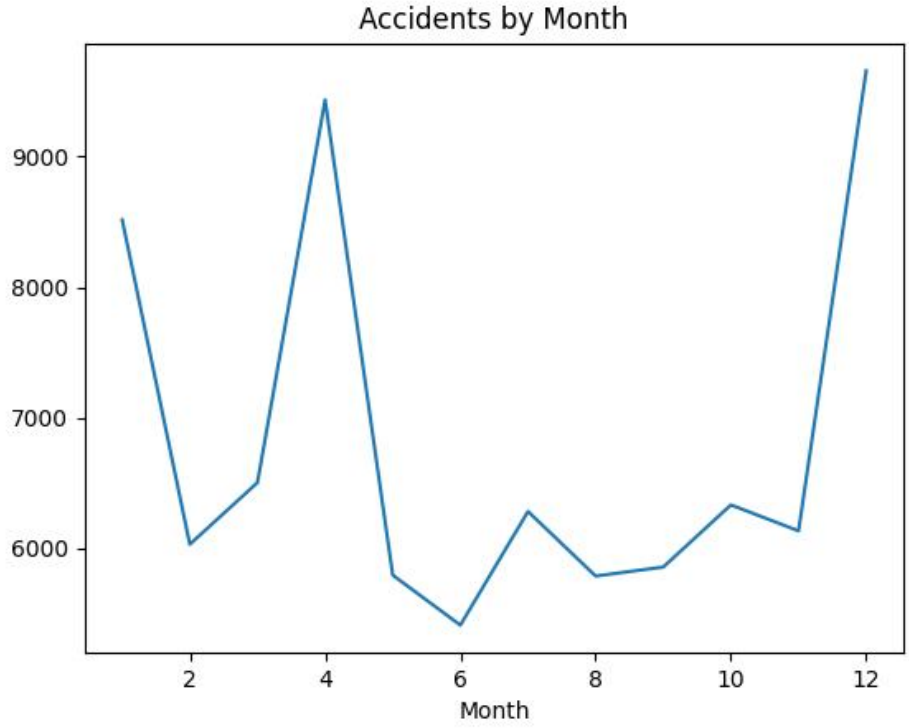
Yearly Accident Distribution

Interpretation: Accidents vary across years, suggesting external influences such as policy or population growth.



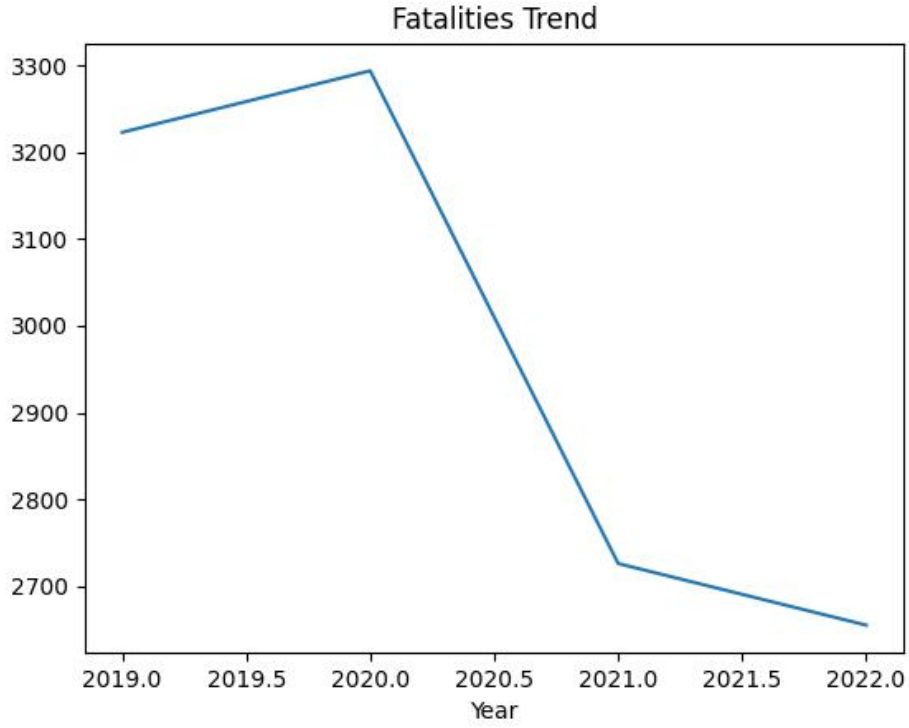
Monthly Accident Trends

Interpretation: Certain months show higher accident rates, indicating seasonal patterns.



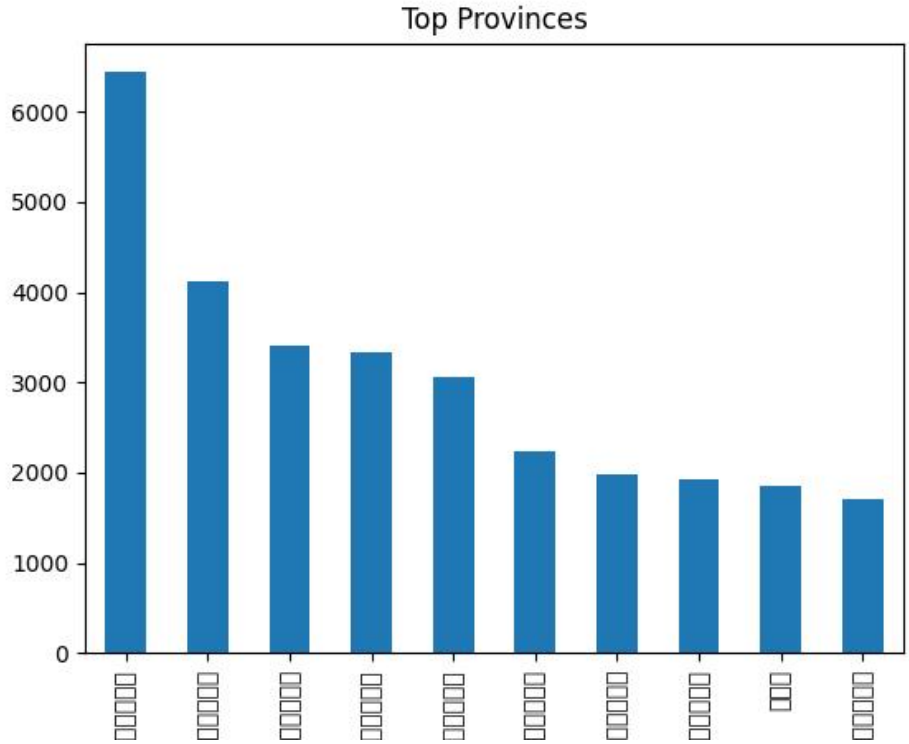
Fatality Trends Over Time

Interpretation: Fatality trends highlight periods of increased road risk.



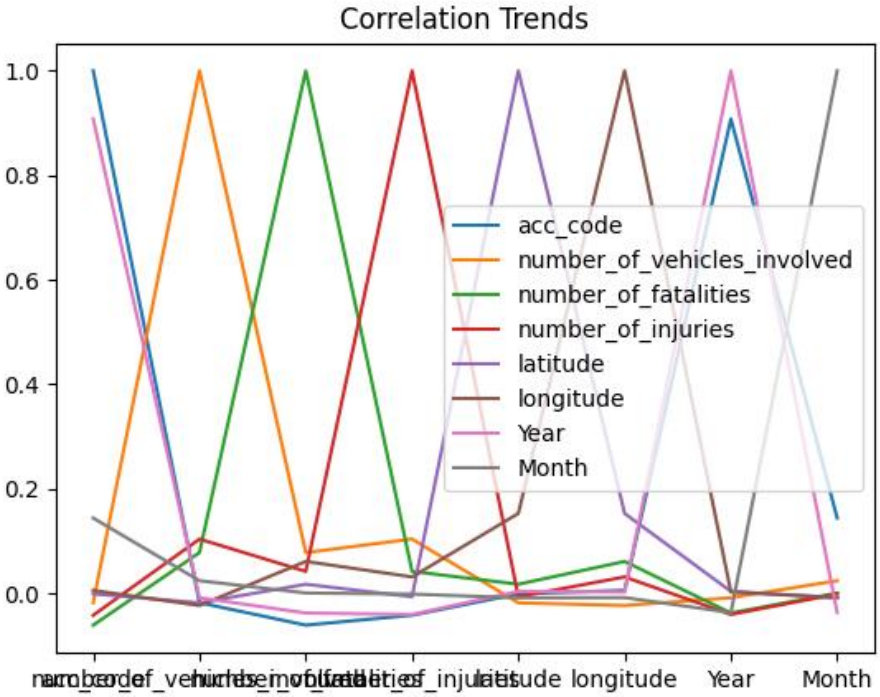
Top High-Risk Provinces

Interpretation: Some provinces consistently record higher accidents due to urban density.



Correlation Analysis

Interpretation: Correlation reveals relationships between numerical variables in the dataset.


Correlation Trends
1.0
0.8
0.6
0.4
0.2
0.0
acc_code
number_of_vehicles_involved
number_of_fatalities
number_of_injuries
latitude
longitude
Year
Month